SCS Winners

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'rvest'


The following object is masked from 'package:readr':

    guess_encoding
```

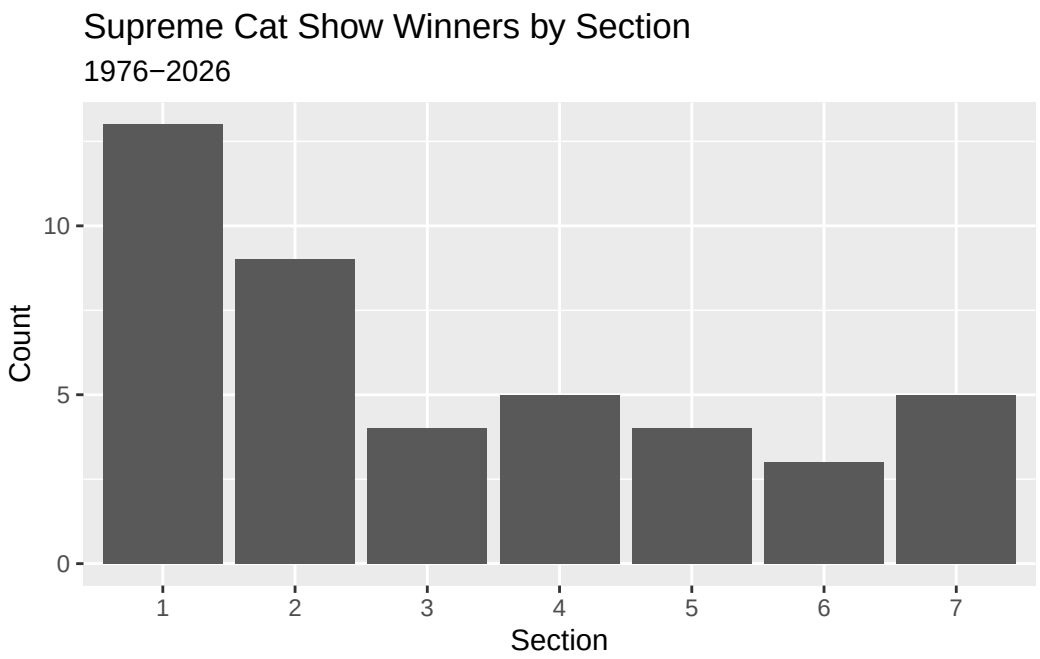
```r
# scrape from webpage
page <- read_html("https://en.wikipedia.org/wiki/Supreme_Cat_Show")
scs_table_raw <- page |> html_elements(".wikitable") |> html_table()
scs_table_raw <- scs_table_raw[[1]]

# start separating the Breed column into breed, sex, and coat (color/pattern)
scs_winners <- scs_table_raw |>
  mutate(sex = str_extract(scs_table_raw$Breed, "male( [a-z]+)?|female( [a-z]+)?")) |>
  mutate(sex = replace_na(sex, "unknown")) |>
  mutate(Breed = str_remove_all(Breed, sex)) |>
  mutate(breed = str_extract(scs_table_raw$Breed, scs_table_raw$Section)) |>
  mutate(breed = replace_na(breed, "unknown")) |>
  mutate(coat = str_remove_all(Breed, breed)) |>
  mutate(coat = str_extract_all(Breed, "(Red|Brown|Chocolate|Seal|Auburn|Cream|Silver|Lilac|
```

```r
# string operations turned my things into lists, I am undoing that
for (i in 1:43) {
  if (length(scs_winners$coat[[i]]) == 0)
    scs_winners$coat[[i]] <- "unknown"
  else if (length(scs_winners$coat[[i]] > 1))
    scs_winners$coat[[i]] <- paste(scs_winners$coat[[i]], collapse = " ")
}

# further refining new columns
scs_winners <- scs_winners |>
  mutate(coat = unlist(coat)) |>
  mutate(Breed = str_remove_all(Breed, coat))

# filling in the rest of the new breed column that was not filled using the information from
for (i in 1:43) {
  if (scs_winners$breed[[i]] == "unknown")
    scs_winners$breed[[i]] <- scs_winners$Breed[[i]]
}

# removing titles from Pedigree Name
scs_winners <- scs_winners |>
  select(-Breed) |>
  rename(name = "Pedigree Name") |>
  mutate(name = str_remove_all(name, ",")) |>
  mutate(name = str_replace_all(name, " and ", " & ")) |>
  mutate(title = str_extract_all(name, "(Grand|Champion|UK|Imperial|Premier|Supreme|Olympian

# fixing lists again
for (i in 1:43) {
  if (length(scs_winners$title[[i]]) == 0)
    scs_winners$title[[i]] <- "none"
  else if (length(scs_winners$title[[i]] > 1))
    scs_winners$title[[i]] <- paste(scs_winners$title[[i]], collapse = " ")
}

# removing titles from names & separating names of owners when there are multiple
scs_winners <- scs_winners |>
  mutate(title = unlist(title)) |>
  mutate(name = str_remove_all(name, title)) |>
  mutate(Owner = str_split(Owner, " & ")) |>
  mutate(owner_1 = "none") |>
  mutate(owner_2 = "none")
```

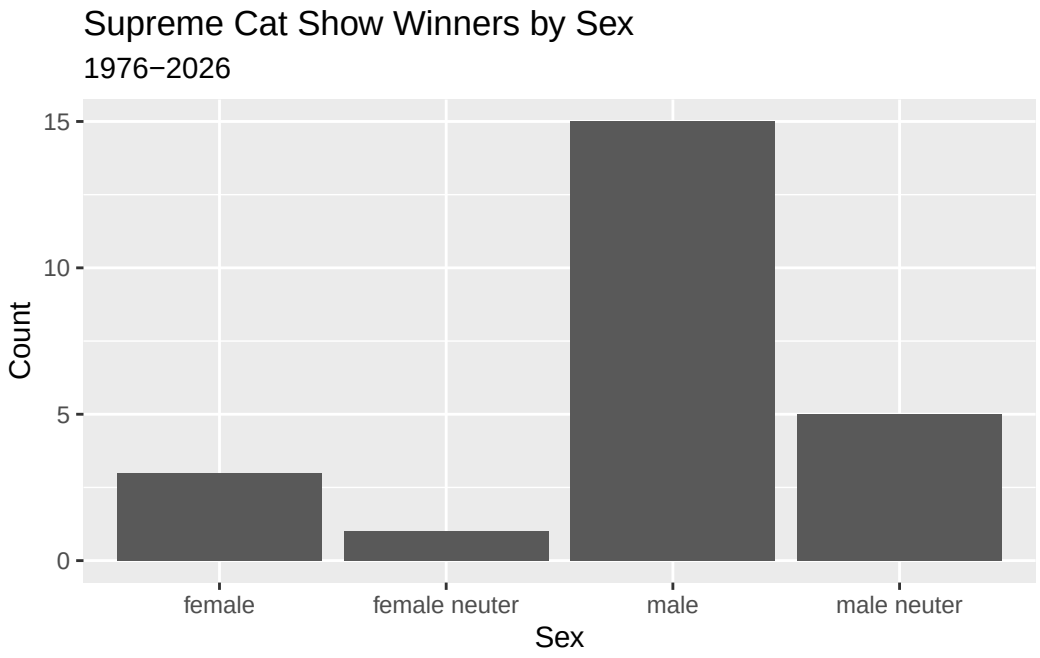
```r
# making sure owners names are complete
for (i in 1:43) {
  if (length(scs_winners$Owner[[i]]) == 1) {
    scs_winners$owner_1[[i]] <- unlist(scs_winners$Owner[[i]])
  }
  else if (length(scs_winners$title[[i]] > 1)) {
    scs_winners$owner_1[[i]] <- scs_winners$Owner[[i]][[1]]
    scs_winners$owner_2[[i]] <- scs_winners$Owner[[i]][[2]]
  }
  if (scs_winners$owner_2[[i]] != "none" && str_detect(scs_winners$owner_1[[i]], " ") == FALS
    scs_winners$owner_1[[i]] <- paste(scs_winners$owner_1[[i]], str_remove(scs_winners$owner_
  }
}

# same operations on Breeders as owners
scs_winners <- scs_winners |>
  mutate(Breeder = str_split(Breeder, " & ")) |>
  mutate(breeder_1 = "placeholder") |>
  mutate(breeder_2 = "placeholder")

for (i in 1:43) {
  if (length(scs_winners$Breeder[[i]]) == 1) {
    scs_winners$breeder_1[[i]] <- unlist(scs_winners$Breeder[[i]])
  }
  else if (length(scs_winners$Breeder[[i]] > 1)) {
    scs_winners$breeder_1[[i]] <- scs_winners$Breeder[[i]][[1]]
    scs_winners$breeder_2[[i]] <- scs_winners$Breeder[[i]][[2]]
  }
  if (scs_winners$breeder_2[[i]] != "none" && str_detect(scs_winners$breeder_1[[i]], " ") ==
    scs_winners$breeder_1[[i]] <- paste(scs_winners$breeder_1[[i]], str_remove(scs_winners$br
  }
}

# removing old columns I'm done with & making a column for whether the breeder and owner are
scs_winners <- scs_winners |>
  select(-c(Breeder, Owner)) |>
  mutate(breeder_owner_match = FALSE)

for (i in 1:43) {
  if (scs_winners$owner_1[[i]] == scs_winners$breeder_1[[i]] || scs_winners$owner_2[[i]] == s
    scs_winners$breeder_owner_match[[i]] <- TRUE
  }
```

```
  if (scs_winners$breeder_2[[i]] == "placeholder") {
    scs_winners$breeder_2[[i]] <- "none"
  }
}

# section numbers for those that are already numbered
scs_winners <- scs_winners |>
  mutate(section = str_extract_all(Section, "[0-9]"))

# assigning numbers to those that are not
for (i in 1:43) {
  if (length(scs_winners$section[[i]]) == 0) {
    if (scs_winners$Section[[i]] == "Persian") {
      scs_winners$section[[i]] <- 1
    }
    else if (scs_winners$Section[[i]] == "Semi-Longhair") {
      scs_winners$section[[i]] <- 2
    }
    else if (scs_winners$Section[[i]] == "British") {
      scs_winners$section[[i]] <- 3
    }
    else if (scs_winners$Section[[i]] == "Foreign") {
      scs_winners$section[[i]] <- 4
    }
    else if (scs_winners$Section[[i]] == "Burmese") {
      scs_winners$section[[i]] <- 5
    }
    else if (scs_winners$Section[[i]] == "Oriental") {
      scs_winners$section[[i]] <- 6
    }
    else if (scs_winners$Section[[i]] == "Siamese") {
      scs_winners$section[[i]] <- 7
    }
  }
  else if (length(scs_winners$coat[[i]] == 1))
    scs_winners$section[[i]] <- unlist(scs_winners$section[[i]])
}

scs_winners <- scs_winners |>
  mutate(section = unlist(section)) |>
  select(-Section) |>
  rename(year = Year)
```

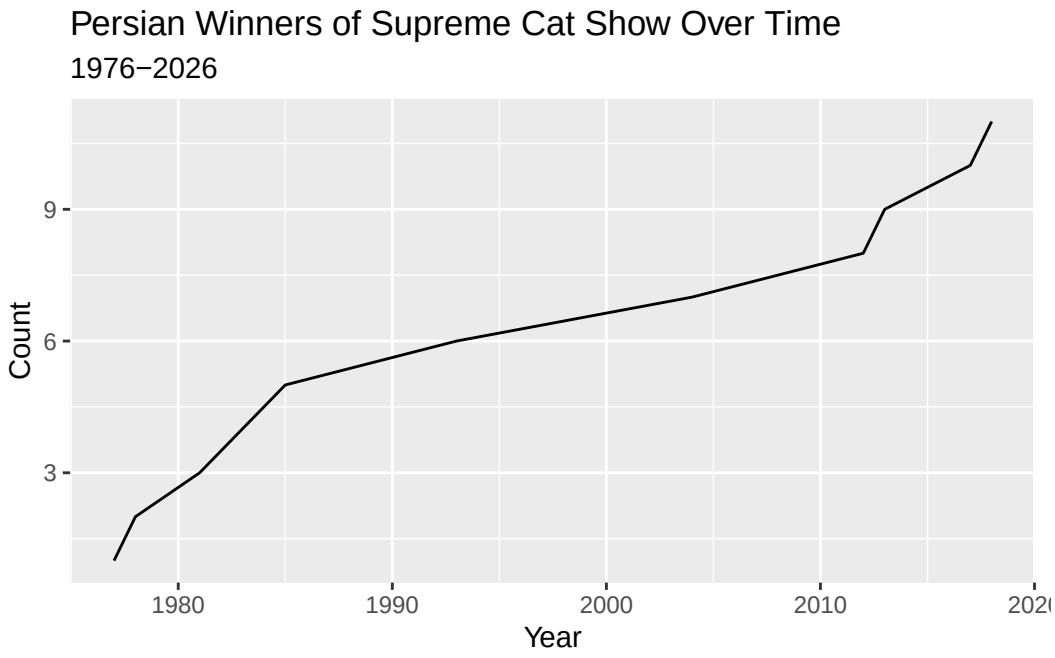
```r
# plots
ggplot(scs_winners, mapping = aes(x = section)) +
  geom_bar() +
  labs(x = "Section", y = "Count", title = "Supreme Cat Show Winners by Section", subtitle =
```

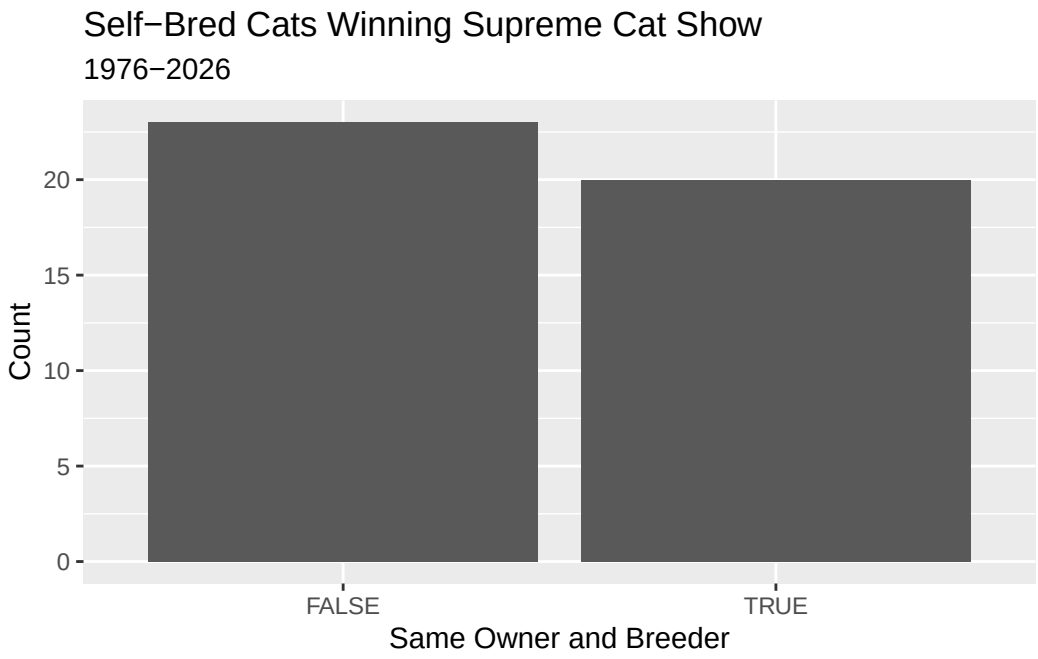



```r
scs_winners |>
  filter(str_detect(sex, "male|female")) |>
  ggplot(mapping = aes(x = sex)) +
  geom_bar() +
  labs(x = "Sex", y = "Count", title = "Supreme Cat Show Winners by Sex", subtitle = "1976-2
```




```
scs_winners |>
  filter(str_detect(breed, "Persian")) |>
  arrange(year) |>
  mutate(ccount = row_number()) |>
  ggplot(mapping = aes(x = year, y = ccount)) +
  geom_line() +
  labs(x = "Year", y = "Count", title = "Persian Winners of Supreme Cat Show Over Time", subt
```




```
scs_winners |>
  ggplot(mapping = aes(x = breeder_owner_match)) +
  geom_bar() +
  labs(x = "Same Owner and Breeder", y = "Count", title = "Self-Bred Cats Winning Supreme Cat
```